

ALICE BALL

American chemist
1892–1916



Alice began her career investigating the chemistry of kava, a plant with a long history in traditional medicine.

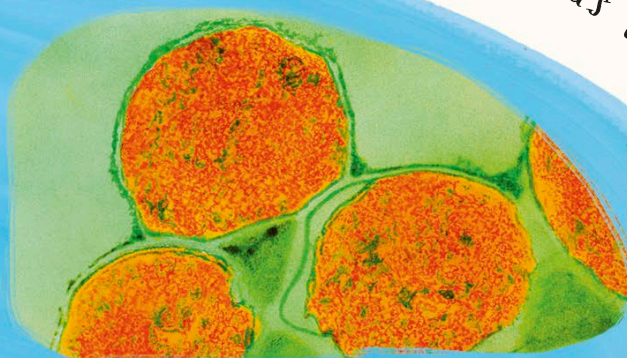
Although Alice Ball died young, she was able to improve life for thousands of other people. Alice spent part of her childhood in the island state of Hawaii, and she decided to return there to study science. In 1915, Alice became the first woman—and the first African American—to receive a master's degree in Hawaii.

Alice was brilliant at unlocking the chemistry of plants—figuring out which substances made certain plants useful as medicines. A local doctor came to her with a problem. Chaulmoogra oil, from the fruits of local trees, was the only treatment he had for a devastating skin illness called Hansen's disease, or leprosy. However, injecting the oil into a patient's bloodstream was painful, because oils and water don't mix, and blood is mainly water. It caused some nasty side effects in his patients.

Alice's breakthrough made life better for thousands of people.

Hansen's disease

Hansen's disease is caused by a certain type of bacteria, called *Mycobacterium leprae*. This disease is also known as leprosy. If it is not treated, leprosy can cause painful sores, damage the eyes, and reduce mobility. Today, around 200,000 new cases of the disease occur each year.



*This is what a cross-section of *Mycobacterium leprae* looks like under a microscope.*

